GRAVITYSORT · CALCULATION NOTE

GravitySort sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade, STEM fixes under Amish's standing instruction of 2026-10-23 (GVS-DEC-001 F1 to F4, F6) and the independent review (climb from the union to the rings, critical speed with frame compliance, shell buckling, tank sway, bounding containment case): header drum drawn at its real 560 mm height (tipping 13.1 degrees), 3/4 in valve and rotameter (union 12.6 kPa, lowest ring 10.0 kPa net at 600 rpm), screen filter on the fluidization hose (loss and cleaning point in section 6), buffer rating 800 N in the BOM, gold balance on 7.75 h of feed; containment of a liner fragment calculated (section 10, PR-016); mass 99.6 kg; cost $594 (USD 139 over the target); PR numbers of GVS-PRS-001 added
0.8	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the design: 1.7 m braced tank post on the centre line (head at the union 8.6 kPa against 8.1 kPa; post, braces and tipping check added to section 11), hose length from the model route,

			flush container with hasp; mass 99.3 kg; cost $580 (USD 125 over the target)
0.7	2026-10-02	Amish Chadha	R2 restated (met on paper); fluidization head and site water decided on 2026-10-02 (GVS-DEC-001); summary corrected on the 1.25 m post; no figure changed
0.6	2026-10-01	Amish Chadha	Table bump stop (GVS-DDR-003 A2) sized in section 9 (impact energy, buffer, setting and asymmetry); mass 96.5 kg; cost $569 (USD 114 over the target)
0.5	2026-10-01	Amish Chadha	R11 restated by Amish (total 100 kg or less, loads 30 kg or less; GVS-DDR-003, A1); R11 met on paper at 95.7 kg
0.4	2026-10-01	Amish Chadha	Constructable design (GVS-DDR-003). Hollow spindle and new bearing positions; mass 95.7 kg (R11 not met, every load under 30 kg); cost $558 reported against the $455 value-engineering target
0.3	2026-09-26	Amish Chadha	Budget approved by Amish ($455); R9 from not met to met on paper
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Budget $450; R12 reworded; frame, pedal outrigger and table stand in 25 x 25 x 1.5 mm tube with a frame member check; mass, cost and results updated
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (speed and G, throughput, water, settling, fluidization supply, power, mass balance, table drive, rotor safety and brake, mass, cost, gold balance)

GravitySort sizing calculations

Version 0.9 (2026-10-25). The kit 1.9.1 upgrade ran the STEM check (GVS-STM-001) over this note, and four details failed and were fixed under Amish's standing instruction of 2026-10-23 (GVS-DEC-001 F1 to F4): the 60 L header drum was drawn only 400 mm tall (45 L) and is now drawn at its real 560 mm, which raises the mid-tank level by 80 mm; a 3/4 in screen filter now protects the 1.0 mm fluidization holes from silt in the recirculated water, at a cost of 0.75 kPa when clean; the bump stop buffer is rated 800 N in the BOM, as section 9 chose; and the gold balance uses the 7.75 h of feed that R2 as restated implies. The independent review of 2026-10-25 (GVS-STM-001) then found that the supply calculation left out the climb of the water from the union to the rings, 4.5 to 5.6 kPa; with it the lowest ring had only 6.0 kPa net at 600 rpm against the proposed 10 kPa. That was fixed under the same instruction (F6) by giving the ball valve and rotameter 3/4 in bores, which cuts the fittings loss from 7.5 to 3.4 kPa: the lowest ring now has 10.0 kPa net at mid-tank and 7.0 kPa with the tank nearly empty and the filter due for flushing. The review also added three checks: the critical speed with the frame's compliance (1,274 to 1,976 rpm, to confirm on the first frame), shell buckling with the bowl empty (safety factor 49) and the sway of the full tank on its post (about 3.5 Hz, near the table drive at 4.0 to 5.0 Hz: a decision for Amish, GVS-DEC-001 O3). The machine now stands on 13.1 degrees with a full tank (criterion 10 degrees) and is 2.26 m tall. Containment of a liner fragment is now calculated (section 10, PR-016): contained if the fragment strikes face first, not shown if it strikes edge first, so it stays at risk and goes to an impact test and a decision for Amish (GVS-DEC-001 O2). Mass 99.6 kg (0.4 kg under PR-011), cost $594 (USD 139 over the value-engineering target). The R numbers below are the "Was" numbers of the product requirements PR-001 to PR-014 in GVS-PRS-001; PR-015 (tipping) and PR-016 (containment) are new. The paragraph that follows is the v0.8 summary, with its figures updated.

On paper, GravitySort meets nine of its fourteen requirements, by calculation or design review, and misses none. Since version 0.4 the note follows the constructable design of GVS-DDR-003, in which every part of the model can be made and fixed to the parts next to it. That design is heavier and dearer than the concept: with the table bump stop (v0.6) and the decisions of 2026-10-02 (v0.8: a 1.7 m braced header tank post and a flush container with a padlock hasp) it weighs 99.6 kg in six loads, with every load under 30 kg (the heaviest is 27.3 kg), which meets R11 as restated by Amish on 2026-10-01 (100 kg or less in all; it was 80 kg) with only 0.4 kg to spare; and the parts cost is $594 against a value-engineering target of $455, USD 139 over (the budget is a hypothetical control target, not a limit; Amish, 2026-10-01). R5 (table ratio) and R6 (pedal power at 80 G) are at risk; R2 (throughput per shift) was at risk until Amish restated it on 2026-10-02 as 200 kg/h of feed time (GVS-DEC-001), which it meets on paper. R4 (recovery) and R14 (liner life) cannot be verified before testing. The bowl, jacket and drive did not change, so speed, G, power, water and settling figures are as in v0.3; the higher head raises the jacket pressure a little, so the lowest burst safety factor at the sprint speed is 16 (v0.7: 17). The spindle is now a 25 x 2 mm tube that carries the fluidization water; its first critical speed falls to about 2,710 rpm, still 2.3 times the sprint speed. The table bump stop decided by Amish on 2026-10-01 (GVS-DDR-003, A2) is sized in section 9: set 3 mm short of the full forward travel, it stops the deck at 2.1 G against 0.6 G at the head end, from 0.24 J per stroke, with a 40 x 30 mm rubber buffer compressed 11 %. The fluidization supply needs a 3/4 in hose. On 2026-10-02 Amish decided a 1.7 m post, braced so a full tank cannot tip it, for the head, and gravity supply from upstream as the site rule, with a treadle, hand or 12 V pump where that is impossible (GVS-DEC-001). With the tank on that post, the 560 mm drum, the 3/4 in valve and rotameter and the filter, the union has 12.6 kPa at mid-tank and the lowest ring 10.0 kPa net at 600 rpm, the proposed target (section 6), so the supply works without a pump on the fluidization line (section 6). To keep the taller post from making the machine easier to tip, the tank now stands on the centre line instead of 180 mm behind it; with a full tank the machine stands on a 13.1 degree slope and the post and braces stay well within yield (section 11). The site still has to bring water to a tank top 2.26 m above the ground.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates for a paper design; nothing is measured.

1. Assumptions

Table 1. Inputs. All are assumptions.

Input	Value	Basis
Feed	200 kg/h dry solids below 2 mm, 30 % solids by mass; 8 h shift; flush every 2 h, 5 min per flush	GVS-REQ-001 R2 and R13
Fluidization water	12 L/min	Mid-range of 8 to 15 L/min (GVS-PRC-001)
Bowl geometry	Liner surface 220 mm lip, 130 mm base, 180 mm deep; four rings 6 mm thick, 12 mm deep, 38 mm pitch; 8 mm PU liner on a 4 mm GFRP shell; 12 mm jacket gap, 4 mm jacket wall	`cad/src/model.py` PARAMS
Drive	48T chainring, 12T freewheel, 1:1 bevel box, 300 mm to 100 mm V-belt: 12:1	`cad/src/model.py` PARAMS
Drivetrain efficiency	Chain 0.97, jackshaft bearings 0.99, bevel box 0.93, V-belt 0.94: 0.839 overall	Typical values for used parts
Spindle drag	0.03 N m per contact-sealed bearing unit, 0.10 N m for the rotary union seal, 1 W windage and splash at 730 rpm	Assumed; no datasheet chosen
Motor option	MotionCore reference 250 W geared hub motor, no-load 250 rpm at full pack voltage, 85 % of that under load; motor and controller 70 % efficient at about 60 W	MTC-PRC-001; motor speed assumed
Bed and film	Bed 12 mm deep when full, 50 % solids of 4,000 kg/m3 (2,500 kg/m3 bulk); 60 % fill at flush, 2.5 kg/L wet; slurry film 5 mm	Assumed; heavy minerals such as magnetite and pyrite dominate the bed
Gold shape	Flaky gold settles at half the velocity of a sphere of the same sieve size	Assumed shape factor
Fluidization supply	Tank water level 1.98 m at mid-tank (1.75 to 2.21 m from nearly empty to nearly full; 60 L drum 380 mm across and 560 mm tall, GVS-DEC-001 F1), union 0.15 m above ground; 2.8 m of 3/4 in hose (the model's route of 2.31 m plus 0.5 m of slack; v0.7 took 4 m), friction factor 0.03; union K = 2 on its 1/2 in bore, ball valve and rotameter K = 4 on a 3/4 in bore (F6; v0.8 took K = 6 for all three on 1/2 in); climb from the union to each ring counted (v0.9); screen filter K = 3 clean and K = 6 at its cleaning point on a 19 mm bore (F2); hole discharge coefficient 0.62	Model layout (1.7 m post, GVS-DEC-001); typical hydraulics; filter K assumed (no datasheet loss curve)
Rotor materials	PU 1,150 kg/m3, 20 MPa tensile; hand-laid GFRP 1,800 kg/m3, 80 MPa tensile; stainless spindle tube 25 x 2 mm, E = 193 GPa	Typical values; spindle per GVS-DDR-003
Brake	Bicycle mechanical disc, 160 mm rotor, pad radius 70 mm, friction 0.4, 150 N per pad with a light pull	Typical values
Table bump stop	17 kg moving on the table; plywood legs E = 7 GPa; rubber buffer about 55 Shore A, E = 3.3 MPa; legs press the deck on the buffer from 5 mm; settings 2 to 6 mm, starting at 3 mm	Typical values; GVS-DDR-003, A2
Gold balance	5 g/t reference ore; oversize loss 5 %, bowl 71 %, table 93 %, smelting 96 %	GVS-PRC-001 Figure 2; assumptions, not calculations
Frame tube	25 x 25 x 1.5 mm mild steel square tube, 1.11 kg/m, yield 235 MPa, E = 200 GPa (was 30 x 30 x 2 mm, 1.76 kg/m)	GVS-DDR-002 (item 10)

Input	Value	Basis
Header tank post and tipping	Full tank 64 kg (60 L of water, drum, valve and strap) at 1.98 m; frame, drive and bowl (loads 1 to 3, 60.3 kg) with their centre of mass 0.55 m up; criterion: the machine stands on a 10 degree slope in any direction with a full tank	Assumed; the criterion is this note's reading of "braced so a full 60 L tank cannot tip it" (GVS-DEC-001)
Value-engineering target	`budget_usd` $455 (was $450, and $350 before that), a hypothetical control target, not a limit	`project.yaml`, GVS-DDR-002 (item 1; $455 approved by Amish on 2026-09-26); Amish, 2026-10-01

2. Speed and G (R3)

The pedal drive gives 12:1, so a 61 rpm cadence turns the bowl at 730 rpm: **59.6 G at 100 mm radius**. The R3 range of 40 to 80 G at 100 mm needs 598 to 846 rpm, a cadence of 50 to 71 rpm; the comfortable 55 to 70 rpm band gives 660 to 840 rpm (49 to 79 G). At 730 rpm the four riffle rings run at 46, 52, 57 and 63 G (wall radii 77.5, 87.0, 96.5 and 106.0 mm), and the lip speed is 8.41 m/s.

R3 asks for the speed to be shown to the operator. A wired bicycle computer (BOM item 19) with its magnet on the spindle pulley and its wheel size set to 1,667 mm reads km/h equal to bowl rpm divided by 10, so 730 rpm reads 73.0. **R3 is met on paper.**

3. Throughput (R2)

At 200 kg/h the bowl is flushed every 2 h, so an 8 h shift has three mid-shift stops of 5 min and 7.75 h of feed time: **1.55 t per shift**. Meeting the old target of 1.6 t would need 206 kg/h, or an 8.25 h shift. Amish restated R2 on 2026-10-02 as 200 kg/h of feed time, about 1.55 t per eight-hour shift with three flush stops (GVS-DEC-001), so **R2 is met on paper**. The bowl's hydraulic capacity at 200 kg/h is supported by the settling check in section 5 but is not proven.

4. Water (R8)

Slurry at 30 % solids carries 0.467 m3/h of water, and fluidization adds 0.72 m3/h: **1.19 m3/h**, 9.2 m3 per shift, against 1.5 m3/h. The slurry is 0.542 m3/h at 1,230 kg/m3, and 0.351 L/s leaves over the lip. **R8 is met on paper.**

The 60 L header tank holds only 3.0 min at full flow, so water must reach it continuously. Its top is now 2.26 m above the ground (v0.8 drew the drum 400 mm tall, 45 L; GVS-DEC-001 F1); lifting 1.19 m3/h by about 2.3 m takes 7.3 W of hydraulic power (v0.8: 6.8 W to 2.1 m; v0.7: 5.5 W to 1.7 m). Amish decided on 2026-10-02 that gravity supply from upstream is the site rule for the first field trial, with a bought treadle or hand pump worked in turns, or a 12 V pump at sites with the MotionCore battery, where that is impossible (GVS-DEC-001). An upstream source must stand more than about 2.3 m above the machine's feet. No pump is in the BOM.

5. Settling in the bowl (R4 plausibility)

This check shows only that the bowl is not too short for fine gold to reach the wall; whether the bed keeps it is the question that only testing answers. At 730 rpm and 100 mm the acceleration is 584 m/s2. The wetted wall is 186 mm long; with an assumed 5 mm film the mean film velocity is 0.091 m/s and the transit time about 2.0 s.

Table 2. Settling velocity at 60 G (Schiller and Naumann drag) and the ratio of transit time to the time to cross the film.

Particle	Settling velocity	Time to cross 5 mm	Ratio
Gold, 20 µm flake	88 mm/s	57 ms	36
Gold, 38 µm flake	213 mm/s	23 ms	87
Gold, 38 µm sphere	426 mm/s	12 ms	173
Gold, 75 µm flake	450 mm/s	11 ms	183
Quartz, 38 µm	61 mm/s	82 ms	25
Quartz, 150 µm	362 mm/s	14 ms	147

Every particle, quartz included, reaches the wall many times over. Separation therefore depends on the fluidized bed letting dense grains displace light ones in the rings, not on reaching the wall. The fluidization upflow of 2.7 mm/s through 0.074 m2 of groove is 4 % of the settling velocity of 38 µm quartz, so it loosens the bed without washing out fine particles. **R4 cannot be verified at TRL 3 and remains the main technical risk.**

6. Fluidization jacket and supply (R4, R8)

The jacket rotates with the bowl, so the water in it gains centrifugal pressure $\rho\omega^2 r^2/2$ on its way out from the rotary union at the axis. Against it, each ring's loaded bed and film push back with a pressure that also rises with ω^2. At 730 rpm with a full bed and the tank half full, the net pressure across a hole is 11.3, 13.9, 17.0 and 20.6 kPa at rings 1 to 4, falling to 10.4 to 16.7 kPa at 600 rpm. It stays positive, so water always flows into the bed; the lowest ring gets 0.74 times the flow per hole of the top ring (v0.7, 1.25 m post: 6.5 to 15.7 kPa and 0.64).

With the valve fully open, 12 L/min now needs only about 74 holes of 1.0 mm. The **89 graded holes of 1.0 mm** decided under GVS-DDR-002 (item 11) are kept: with them the valve is set slightly closed at mid-tank, which leaves margin as the tank level falls, and grading the holes toward the lower rings evens out the flow.

The supply needs a 3/4 in hose. With the tank on the 1.7 m post decided by Amish on 2026-10-02 (GVS-DEC-001) and the 60 L drum drawn at its real 560 mm height (F1), the water level is 1.98 m at mid-tank, which gives 18.0 kPa of static head at the union; the 3/4 in hose loses 1.1 kPa and the union, the 3/4 in valve and rotameter (F6) and the clean screen filter 4.2 kPa (the filter 0.75 kPa of it), leaving **12.6 kPa at the union**. The water then climbs 0.46 to 0.57 m up the spindle and jacket to the rings, which costs 4.5 to 5.6 kPa, while the turning jacket adds $\rho\omega^2 r^2/2$. The lowest ring (ring 1, the smallest radius) gets **10.0 kPa net at 600 rpm** at mid-tank, just the proposed 10 kPa; the union needs 12.7 kPa, a water level of 1.98 m, for it. From nearly full to nearly empty the union sees 14.9 to 10.4 kPa; at the lowest level the net pressure at the lowest ring at 600 rpm is still 7.7 kPa, under the target but positive, so water still enters every ring. Keeping the tank above half full (the gravity supply from upstream runs continuously) holds the target. With a 1/2 in hose the hose loss would be 8.2 kPa and the union would see only 5.5 kPa.

v0.8 had no climb term and took K = 6 on a 1/2 in bore for the union, valve and rotameter (7.5 kPa); with the climb, those fittings would have left the lowest ring 6.0 kPa net at mid-tank. The independent review found the missing climb; the fix (F6, GVS-DEC-001) keeps the 1.7 m post and the no-pump rule that Amish decided on 2026-10-02.

Fluidization against the bed. The upflow of 2.7 mm/s through the ring grooves is about the minimum fluidization velocity of 75 um quartz at 60 G (2.9 mm/s by the Wen and Yu correlation) and well below that of coarser or denser grains (11.5 mm/s for 150 um quartz, 27 mm/s for 150 um pyrite), so most of the bed is loosened, not fully fluidized. Commercial fluidized bowls also run below full fluidization of their coarse fraction and their recovery is found by test; the iCON i150 uses 10 to 30 L/min at 35 to 100 kPa on a smaller bowl, against 12 L/min at about 10 kPa here. This is why PR-004 is at risk, and the TRL 4 recovery test should sweep the fluidization flow (GVS-DEC-001 O5). With the 1.25 m post the union had 3.7 kPa (v0.7). No pump is needed on the fluidization line.

Screen filter (GVS-DEC-001 F2, PR-008). Commercial fluidized concentrators need clean fluidization water (Veiga and Gunson 2020), and GravitySort runs on recirculated pond water through 1.0 mm holes. A 3/4 in screen filter with a 120 mesh (0.125 mm) stainless screen and a flush valve now sits on the hose's drop below the rotameter, stopping any grain larger than an eighth of the hole size. Clean it is taken as K = 3 on the 19 mm bore, 0.75 kPa at 12 L/min; it is flushed when it reaches about twice that (K = 6, 1.49 kPa), which the operator sees as the rotameter flow falling. With the tank nearly empty and the filter at its cleaning point the lowest net ring pressure at 600 rpm is 7.0 kPa, still positive. The filter's real loss curve is not known; the K values are assumptions to check against the datasheet.

7. Power (R6, R7)

The power to bring the slurry and fluidization water (0.385 kg/s) up to lip speed is $m \cdot v^2$, an upper bound that counts the slip loss. The drag from the two bearing units and the union seal is added, then divided by the drivetrain efficiency of 0.839.

Table 3. Power at the pedals.

Bowl speed	Slurry and water	Drag	At the pedals
600 rpm (40 G)	18.4 W	10.6 W	34.6 W
730 rpm (60 G)	27.2 W	13.2 W	**48.2 W**
850 rpm (80 G)	36.9 W	15.8 W	**62.8 W**
1,200 rpm (100 rpm sprint)	73.6 W	24.5 W	116.9 W

At 60 G the rider supplies 48.2 W with a crank torque of 7.6 N m, inside the 60 W target. At 80 G the target is exceeded. The union seal drag is a guess and makes up most of the drag. **R6 is at risk** (met at 60 G, not at 80 G). The TRL 2 estimate of about 45 W is corrected to 48.2 W.

With the motor, the 250 W MotionCore reference drive has a margin of **5.2 times** at 60 G and 4.0 times at 80 G, so **R7 is met on paper**. At 70 % motor and controller efficiency a shift takes **0.55 kWh** from the pack (TRL 2: about 0.5 kWh). A motor chain step-up of 1.20:1 means the motor's 250 rpm no-load speed gives at most 900 rpm at the bowl, and about 765 rpm under load; the ratio must be set for each motor so that its no-load speed at full pack voltage stays within 900 rpm.

8. Mass balance (R5)

The four ring grooves hold 0.89 L. At 60 % fill and 2.5 kg/L each flush yields **1.33 kg**, a mass pull of **0.33 %** of the 400 kg fed in 2 h, against 0.5 %. Four flushes give 5.3 kg per shift. Reaching 100 g needs a table reduction of **53:1**; a single pass on a small table is unlikely to achieve that, but two passes at about 8:1 each (64:1) would give about 83 g. **R5 is at risk** until the table ratio is shown.

9. Shaking table drive

The table head turns at cadence x 4.48 (4:1 chain, 140 mm take-off to 125 mm head pulley): 246 to 314 strokes/min at 55 to 70 rpm. The 240 to 300 strokes/min range needs a cadence of 54 to 67 rpm. At 270 strokes/min and a 15 mm stroke the deck's peak speed is 0.21 m/s and the head takes about 13 W, about 15 W at the pedals, so table clean-up is light work.

Bump stop (asymmetric stroke, GVS-DDR-003 A2). A plain eccentric on flexure legs moves the deck in a nearly symmetric sine, which does not walk gold along the table. Gold walks toward the concentrate box only if the deck turns back sharply at the end of its forward stroke (away from the head), so the stop sits there: the deck strikes it at speed and the particles slide on. A rigid pitman cannot strike anything (it drags the deck through the whole circle of the eccentric), so the pitman pin works in a slot in the deck cheeks with 8 mm of free play toward the far end. The pin pulls the deck toward the head; the four plywood legs, set leaning so that they press the deck onto the buffer, push it forward against the pin and then into the stop. Once the deck is stopped, the pin runs on into the slot and comes back for it.

The legs (80 x 18 mm marine plywood, 697 mm between cleats, E taken as 7 GPa, clamped at both ends) give 38.6 N/mm for the four, so the deck alone would swing at 7.6 Hz against the 4.5 Hz drive. Because they are stiffer than the deck's inertia at the drive speed (17 kg x 28.3^2 = 13.6 N/mm), the pin pull never falls to zero on the forward stroke (254 to 554 N at the 3 mm setting): the deck follows the pin up to the stop. Set 5 mm past the stop, the legs press the deck on the buffer with 193 N. The buffer is a bonded rubber cylinder 40 mm across and 30 mm long, about 55 Shore A (E about 3.3 MPa, shape factor 0.33): about 169 N/mm.

The setting is the gap between the buffer and where the striker would be at full forward travel: turning the stud in by g shortens the stroke at the forward end only. The deck meets the buffer at $v = \omega \sqrt{(A^2 - (A - g)^2)}$, with A = 7.5 mm and ω = 28.3 rad/s.

Table 4. Bump stop at 270 strokes/min, 17 kg moving.

Setting	Stroke	Strike speed	Impact energy per stroke	Buffer squeeze	Peak force	Stop deceleration	Head-end acceleration	Ratio
2 mm	13 mm	0.14 m/s	0.18 J (0.8 W)	3.0 mm (10 %)	504 N	1.9 G	0.61 G	3.1
3 mm (start)	**12 mm**	**0.17 m/s**	**0.24 J (1.1 W)**	**3.2 mm (11 %)**	**539 N**	**2.1 G**	**0.61 G**	**3.4**
4 mm	11 mm	0.19 m/s	0.30 J (1.3 W)	3.3 mm (11 %)	565 N	2.2 G	0.61 G	3.6
6 mm	9 mm	0.21 m/s	0.37 J (1.7 W)	3.5 mm (12 %)	594 N	2.4 G	0.61 G	3.9

The ratio of the stop's deceleration to the smooth turn at the head end is the asymmetry: about 3 to 4 times over the useful range. The buffer squeeze includes the legs' push and stays at 12 % or less of its length against a usual working limit of 20 %, and the peak force is under 600 N, so a buffer rated 800 N or more in compression is chosen. The energy lost in the stop, 0.8 to 1.7 W, adds little to the 15 W at the pedals. The pin pull is at most 593 N, 5.2 MPa of shear on the 12 mm pin. The stud gives 0 to 8 mm of setting, matching the slot's free play; 3 mm is the starting point, to be tuned in the first table test (TRL 4). The leg stiffness and buffer rate are estimates; the lean of the legs is set by the push on the buffer, not calculated, when the table is built.

10. Rotor safety: inertia, burst, spindle and brake (R12)

Inertia and energy. Integrating the liner, shell, jacket, jacket water, rings and a full bed along the cone gives a rotating group of 6.47 kg with I = 0.0535 kg m2, plus 0.0031 kg m2 reflected from the drive pulley: 0.0566 kg m2 in total. The stored energy is **165 J at 730 rpm**, 224 J at 850 rpm, 252 J at 900 rpm and 447 J at 1,200 rpm (TRL 2: about 150 J at 730 rpm).

Burst. The checks run at 1.2 x 900 = 1,080 rpm and at 1,200 rpm, the speed a rider sprinting at a 100 rpm cadence could reach.

Table 5. Stresses at the lip region.

Item	1,080 rpm	1,200 rpm	Safety factor at 1,200 rpm
Jacket wall hoop (water 111 and 135 kPa)	4.0 MPa	4.9 MPa	16 on 80 MPa GFRP
Bowl shell hoop, jacket empty (75 kPa at 1,200 rpm)	2.1 MPa	2.6 MPa	30
PU liner, unsupported hoop	0.19 MPa	0.24 MPa	Liner is pressed onto the shell
Ring lip bending under the bed	0.32 MPa	0.39 MPa	51 on 20 MPa PU

The margins are large because the bowl is small and slow. They do not cover poor lamination, voids or a liner that debonds, which is why the liner must be checked before every run.

Containment of a liner fragment (PR-016). Amish decided on 2026-10-01 that containment is to be checked by calculation before any spin test, with a steel band round the tub at the bowl's height only if the check fails. A quarter of the liner and shell (0.52 kg) leaving the lip at 15.3 m/s (1,200 rpm) carries 61 J. It leaves on a tangent at radius 122 mm and meets the tub wall (inside radius 209 mm) obliquely, so only the normal share $1 - (122/209)^2 = 0.66$ of its energy loads the wall: 40 J at 1,200 rpm, 22 J at 900 rpm. A first-order plugging model for a ductile plate, $E = \pi\, d\, t^2\, \tau / 2$ with $\tau = 20$ MPa for HDPE and $t = 6$ mm, gives 57 J if the fragment strikes face first (about 50 mm of contact) and 14 J if it strikes edge first (the 12 mm thickness of liner and shell). The model ignores the dishing of the wall, which absorbs much more energy in a ductile plate, so both figures are lower bounds. Face first the tub contains the fragment at both speeds; edge first containment is not shown even at 900 rpm. The independent review called the quarter of liner and shell arbitrary; a quarter of the whole bowl group (1.62 kg, with the jacket, its water and the bed) puts 125 J into the wall at 1,200 rpm, which the tub is not shown to contain either. **PR-016 stays at risk**, and the STEM check records it as failing with a fix proposed (S10). A 3 mm steel band about 200 mm tall would add a plugging energy of about 37 J edge first ($\tau = 216$ MPa), enough with the tub at 1,200 rpm, but it weighs about 6.4 kg and would take the machine to about 106 kg, over PR-011. Because the check neither clearly passes nor clearly fails, and because the band breaks a requirement Amish restated, the choice goes to him (GVS-DEC-001 O2): the recommendation is an impact test of the tub wall as the first TRL 4 item, before any spin test.

Spindle. The spindle is a 25 x 2 mm stainless tube, because the fluidization water runs up its bore (GVS-DDR-003). A 300 N belt pull 55 mm below the centre of the lower bearing gives 21 MPa of bending. With the bowl centre 210 mm above the upper bearing, the first critical speed of the shaft on rigid bearings is about 2,710 rpm, 2.3 times the sprint speed (v0.3, solid shaft with the concept's bearing positions: 4,820 rpm). The bearings, however, hang under pairs of 25 x 25 x 1.5 members 600 mm long, which are about as compliant as the shaft, and the 210 mm overhang over the 165 mm bearing span magnifies their movement at the bowl. Counting that (v0.9, independent review), the first critical speed is **1,274 to 1,976 rpm**, from member ends pinned to fully fixed. It is well above the 900 rpm limit in every case, but above 1.25 times the 1,200 rpm sprint speed only if the welded member ends act nearly fixed. A tap test on the first frame (TRL 4) settles it; if it is under 1,500 rpm the spindle supports should be stiffened (GVS-DEC-001 O4).

Shell buckling. If the holes blocked with the bowl empty, the full jacket would press on the outside of the 4 mm GFRP shell: 139 kPa at 1,200 rpm. Taking the riffle rings at 38 mm pitch as stiffeners and ignoring the liner, the Windenburg and Trilling formula gives a buckling pressure of 6.8 MPa (E = 7 GPa): a safety factor of 49. A 50 g lump of uneven concentrate at 100 mm radius adds a 29 N rotating load at 730 rpm.

Stopping. With the drive stopped, the freewheel lets the bowl coast. From 850 rpm it takes about **20 s** to stop with fluidization water running (the water leaving the lip acts as a brake) and 32 s with the water off, both longer than the 15 s in R12. A bicycle mechanical disc brake on the spindle (BOM item 18) gives 8.4 N m with a light pull and stops the bowl in 0.6 s; only 0.18 N m is needed for a 15 s stop, so the operator should brake gradually. Each stop warms the rotor by 4.1 K. The brake lever has a parking latch, and a pin on the same cable locks one lid clamp, so the lid can be opened only with the brake applied. With the motor, a lid switch on a MotionCore brake input removes torque (MotionCore stop category 0 does not brake, so the mechanical brake is still needed).

Speed limit. The motor case is capped twice: by the 1.20:1 step-up (section 7) and by the MotionCore speed limit, with its sensor on the spindle. The pedal case is not capped by gearing, because 900 rpm needs a cadence of only 75 rpm. R12 as reworded under GVS-DDR-002 (item 9) asks instead for a burst safety factor of 10 or more at the highest reachable pedal speed (1,200 rpm at a 100 rpm cadence) and a speed display. The lowest factor at 1,200 rpm is 16 (jacket wall; 17 in v0.7, before the higher head of the 1.7 m post raised the jacket pressure), and the display is item 19, so **R12 is met on paper**; containment of a liner fragment by the tub and lid is still not verified.

11. Mass and size (R11)

Table 6. Transport loads (estimates), constructable design.

Load	Mass
1. Base frame with bearing plates, tank post and braces, tank cradle, spindle, bearings, brake, union, display (16.4 m of tube at 1.11 kg/m)	27.3 kg (v0.7: 25.2 kg; v0.3: 18.1 kg)
2. Drive and pedal station (outrigger 2.1 m of tube, round seat tube, end plates; guards)	18.9 kg (17.8 kg)
3. Bowl, jacket, hub, tub, standpipe and pipes, lid, hopper and its support	14.2 kg (11.5 kg)
4. Table deck, wash pipe and striker	7.8 kg (7.2 kg)
5. Table base, flexure legs, head plate and bearings, pitman, tensioner, table belt guard, bump stop, launder and box	22.2 kg (15.7 kg)
6. Water tank, valve, rotameter, screen filter and hoses; flush container with its hasp	6.3 kg (v0.8: 6.0 kg)
Hardware	3.0 kg (2.0 kg)
Total	**99.6 kg** (v0.8: 99.3 kg; v0.7: 96.5 kg; v0.3: 77.3 kg)

The constructable design adds 2.7 m of frame tube (lower rails, member pairs either side of the shafts, drop posts, end post), about 10 kg of plate (bearing plates 1.4 kg, tank cradle 3.1 kg, head plate and shelf 3.9 kg, outrigger end plates, caliper bracket, hopper ring) and parts the concept left out (hopper support, standpipe and pipes, tensioner, table belt guard, launder and box legs). The table bump stop (bracket, buffer, stud and striker) adds 0.8 kg (v0.5: 95.7 kg). The 1.7 m tank post and its two braces add 1.9 m of tube, 2.1 kg, and the flush container with its hasp 0.7 kg (v0.8, decisions of 2026-10-02). The motor cradle (1.3 kg) goes with the motor option and is not counted. The screen filter of v0.9 (GVS-DEC-001 F2) adds 0.3 kg. The heaviest load is the frame at 27.3 kg, so every load is under 30 kg, and the total is 0.4 kg under the 100 kg of R11 as restated by Amish on 2026-10-01 (GVS-DDR-003, A1): **R11 is met on paper, with very little margin**. The detailed model (`cad/fidelity.json`) gives a larger figure from its solid volumes, because it counts the motor option and the loose flush container, models bought parts such as the crankset, saddle, pulleys and pillow blocks as solid envelopes, and draws the perforated guards as plain sheet; it is not a weighing either. R11 should be treated as at risk until parts are weighed at TRL 4. The savings of about 16 kg listed in GVS-DDR-003 (A1, option b) are to be tried at TRL 4. The 30 min assembly time is not verified. The overall size from the model is 2.78 x 0.79 x 2.26 m (the tank top is the highest point; v0.8: 2.10 m with the drum drawn too short; v0.7: 1.65 m), with the loose flush container left out.

Frame member check. The most loaded members are the two spindle members, 600 mm long, which carry the lower bearing plate between them; the check takes one member alone. Taking the 300 N belt pull, the weight of the rotating group and spindle (about 93 N) and the 29 N unbalance load together as one central load of 422 N on a pinned span (an upper bound), the 25 x 25 x 1.5 mm tube sees 61 MPa of bending, a safety factor of 3.9 on 235 MPa, and deflects at most 0.73 mm. That is adequate on paper; bearing alignment under the belt pull should be checked on the first frame (TRL 4, on hold).

Header tank post, braces and tipping. The tank post is 995 mm of the frame tube, standing on the tank post member on the machine's centre line, with the cradle 1.7 m above the ground. Two braces of the same tube (717 mm cut length) run from the post, 1,350 mm up, to the front and back top side rails, 250 mm toward the table end, so the post is held both along and across the machine. The check takes the frame, drive and bowl (loads 1 to 3, 60.3 kg, centre of mass taken 0.55 m up) and a full tank (64 kg at 1.98 m with the 560 mm drum): 124.3 kg with its centre of mass 1.29 m up on the centre line. The machine is narrowest across, 600 mm between the outsides of the legs, so it tips most easily sideways.

Table 6a. Tipping with a full tank (frame-borne mass only; the table and the outrigger's own feet are not counted, which is conservative).

Layout	Slope at which it tips sideways	Side push at the tank that tips it on level ground
1.25 m post, tank 180 mm behind the centre line (v0.7)	11.6 degrees	174 N
1.7 m post, tank 180 mm behind the centre line	9.2 degrees	128 N
1.7 m post, tank on the centre line, drum drawn 400 mm tall (v0.8)	13.5 degrees	193 N
1.7 m post, tank on the centre line, 560 mm drum (this design, PR-015)	**13.1 degrees**	**185 N**

Raising the post where it stood would have made the machine tip on a slope of about 9 degrees, under the 10 degree criterion decided by Amish on 2026-10-03; on the centre line it stands on 13.1 degrees with the drum at its real height, better than the old 1.25 m layout. **PR-015 is met on paper.** Along the machine the pedal outrigger and the table end give a much longer base. The post and braces are checked at the largest side push the machine can take before it tips (185 N at the tank): the post above the braces sees 112 MPa of bending (safety factor 2.1 on 235 MPa), and each brace at most 556 N (3.9 MPa, against an Euler buckling load of 44 kN). So the machine would tip before the post or braces yield. The full tank on the post member, taken alone on pinned ends with the braces ignored, gives 83 MPa (safety factor 2.8), and 4.5 MPa of compression in the post. **Sway of the full tank (v0.9, independent review).** Above the brace node the post is a 630 mm cantilever of 25 x 25 x 1.5 tube under the 64 kg tank, which gives a first lateral frequency of about 3.5 Hz. The table is driven at 4.0 to 5.0 Hz from the frame's table end, and the bump stop's impacts add harmonics, so a full tank could sway when the table runs. Raising the brace node to 1.6 m would give about 7.5 Hz, but the longer braces add about 0.5 kg, which takes the machine over the 100 kg of PR-011; running the table with the drum drained is the alternative. This goes to Amish (GVS-DEC-001 O3).

A push of about 185 N at the tank (a person leaning on it, or a hose pulled hard) tips the machine on level ground, so the tank should be filled with the machine standing level and nobody should climb on the frame or hang anything on the tank.

12. Cost (R9)

Value-engineering target: USD 455. Estimated cost of the constructable design: USD 594 (USD 139 over the target), excluding the MotionCore kit and reference motor ($335 per MTC-CAL-001) and the battery. The target is a hypothetical control target that keeps the design on a value-engineering lens, not a spending limit (Amish, 2026-10-01). The concept BOM was $455; the parts added to make the design buildable (GVS-DDR-003) add $103, mainly the table stand and head ($28 to $52: pillow blocks, plywood legs, head plate, tensioner), the frame ($32 to $43: more tube and the plates), the tub fittings ($12 to $26: standpipe, tailings pipe, two rubber grommets) and the hub, bolts and spacers ($10). The table bump stop adds $11 (bracket and striker $6, buffer and fixings $5). The decisions of 2026-10-02 add $11 (GVS-DEC-001): the 1.7 m braced tank post adds about 1.9 m of tube and its welding to the frame ($43 to $47), and the flush container with its padlock hasp is a new line ($7). The screen filter of v0.9 (GVS-DEC-001 F2) adds $14 to line 13 (DIG P13-120, $14.01, cheapsprinklers.com, checked 2026-10-25). R9 is reported as **over the value-engineering target by USD 139**. The savings worth trying are in the Value engineering section of the design decisions register (GVS-DEC-001).

13. Gold balance

On the 5 g/t reference ore, 7.75 g of gold enters per shift at 200 kg/h for the 7.75 h of feed time (v0.8 used the full 8 h, 8.0 g; GVS-DEC-001 F4); 7.4 g passes the screen, 5.2 g is held in the bowl, 4.9 g stays on the table and 4.7 g is smelted: **about 60 % overall**. Losses are 0.4 g to oversize, 2.1 g to bowl tailings, 0.4 g to table tailings (re-run) and 0.2 g to slag. The stage recoveries are assumptions, not results of this note.

14. Results

Table 7. Requirements against the calculations, not met items first.

ID	Requirement	Value	Target	Status
R2	Ore per shift	200 kg/h; 1.55 t with 3 flush stops	200 kg/h of feed time, about 1.55 t per 8 h shift (restated 2026-10-02, GVS-DEC-001)	Met on paper
R5	Concentrate for smelting	Bowl pull 0.33 %; 5.3 kg/day needs 53:1 on the table	0.5 % or less; 100 g or less	At risk (two table passes)
R6	Pedal power	48.2 W at 60 G; 62.8 W at 80 G	60 W or less	At risk (met at 60 G, not at 80 G)
R4	Fine-gold recovery	Settling ratio 36 or more for 20 µm flakes; overall 60 % assumed	80 % and 50 % in the bowl; 60 % overall	Not verifiable at TRL 3 (at risk)
R14	Liner life	No wear data for this PU on quartz	500 h; replace in 30 min	Not verifiable at TRL 3
R1	No mercury	No amalgamation step	No mercury at any step	Met (design review)
R12	Guards, speed limit, stop time	Brake stop 0.6 s (coast 20 s); motor capped; pedal: burst SF 16 at 1,200 rpm with speed display	Guarded; motor 900 rpm or less; pedal SF 10 or more at sprint speed with display; stop within 15 s	Met (paper)
PR-016	Containment of a liner fragment	Contained face first (57 J against 40 J at 1,200 rpm); not shown edge first (14 J against 22 J at 900 rpm)	Contained at 1,200 rpm	At risk (impact test; band option to Amish)
PR-015	Stability with a full tank	13.1 degrees; post SF 2.1 at the tipping push	10 degrees or more	Met (paper)
R3	Bowl G and speed display	40 to 81 G over 600 to 850 rpm; display item 19	40 to 80 G, speed shown	Met (paper)
R7	Motor margin	5.2 times at 60 G (4.0 at 80 G)	3 times or more	Met (paper)
R8	Water use and quality	1.19 m3/h; screen filter on the fluidization hose	1.5 m3/h or less; filtered fluidization water	Met (paper)
R10	Local workshop build	Welding, drill press, printed mold and six-segment core; set-screw inserts, taper bush, welded nipple	No lathe	Met (design review); casting route unproven
R11	Transport mass	99.6 kg in 6 loads, heaviest 27.3 kg	100 kg or less; loads 30 kg or less (restated 2026-10-01)	Met on paper; at risk (0.4 kg of margin)
R13	Quick, secure clean-up	Toolless lid clamps; flush container and concentrate box each with a padlock hasp	5 min, no tools	Met (paper); time not verified
R9	Parts cost	$594	Value-engineering target $455	Over the value-engineering target by USD 139

15. Corrections to TRL 2 figures

- Pedal power 45 W to 48.2 W; motor energy 0.5 to 0.55 kWh per shift.
- Stored energy at 730 rpm 150 J to 165 J.
- Machine mass 75 kg to 88 kg (the TRL 2 figure is replaced by a load-by-load estimate); 77.3 kg in v0.2 with the lighter tube.
- Parts cost $437 to $465 (brake and speed display added), then $455 in v0.2 with the lighter tube; MotionCore $325 to $335.
- Throughput per shift 1.6 t to 1.55 t once flush stops are counted.
- Table reduction "about 50:1" to 53:1 needed for 100 g.
- v0.4 (constructable design, GVS-DDR-003): mass 77.3 kg to 95.7 kg; parts cost $455 to $558; spindle first critical speed 4,820 to 2,710 rpm; overall size 2.72 x 0.78 x 1.65 m to 2.78 x 0.79 x 1.65 m.
- v0.6 (table bump stop, GVS-DDR-003 A2): mass 95.7 kg to 96.5 kg; parts cost $558 to $569.
- v0.8 (decisions of 2026-10-02, GVS-DEC-001): union pressure 3.7 to 8.6 kPa at mid-tank; hose taken from the model route (2.8 m, was 4 m); lift to the tank 1.7 to 2.1 m; lowest burst safety factor 17 to 16; mass 96.5 kg to 99.3 kg; parts cost $569 to $580; overall height 1.65 m to 2.10 m.
- v0.9 (STEM fixes, GVS-DEC-001 F1 to F4 and F6, and the independent review): drum 400 to 560 mm tall (45 to 60 L as drawn); mid-tank level 1.90 to 1.98 m; screen filter added (0.75 kPa); climb from the union to the rings counted and the valve and rotameter at 3/4 in: union pressure at mid-tank 8.6 to 12.6 kPa, lowest ring at 600 rpm 10.4 to 10.0 kPa net; first critical speed 2,710 rpm on rigid bearings becomes 1,274 to 1,976 rpm with the frame's compliance; shell buckling and tank sway added; tipping 13.5 to 13.1 degrees (193 to 185 N); post safety factor 2.3 to 2.1; lift to the tank 2.1 to 2.3 m; gold per shift 8.0 to 7.75 g in, 4.8 to 4.7 g smelted; mass 99.3 to 99.6 kg; parts cost $580 to $594; overall height 2.10 to 2.26 m; containment of a liner fragment calculated.

Safety: Every rotor figure here is a paper estimate. The large burst margins assume sound lamination and a bonded liner; a cracked or debonded liner must never be run. The bowl coasts for about 20 s after the drive stops, so the brake must be applied and parked before the lid is opened. A full header tank 1.7 m up makes the machine top-heavy: about 185 N pushed sideways at the tank tips it, so fill it only with the machine standing level and never climb on the frame. Nobody should run the bowl above 900 rpm, even though a rider can reach it; the speed display is the only warning on the pedal drive, and the reworded R12 relies on the structural margin, not on a speed cap.